PART 1: ENTERPRISE DATABASE BLUEPRINT (REFINED VERSION 2.0)

System Title: Secure Examination Content Management System with Integrated Peer-Review Pipeline

Institution: Lagos State University (LASU) — Faculty of Computing and Information Technology (FCIT)

Target RDBMS: MySQL 8.0+ / InnoDB Storage Engine (Fully 3NF Normalized Architecture)

1. ARCHITECTURAL OVERVIEW & ENTERPRISE DESIGN STRATEGIES

This Enterprise Database Blueprint defines a production-ready, highly secure, fully 3NF-normalized database architecture tailored for the Faculty of Computing and Information Technology (FCIT) at Lagos State University (LASU).

Key Enterprise Strategies & Architectural Guarantees:

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1. Soft Delete Strategy (Data Preservation & Recovery):

Important entities (users, departments, courses, examination_papers) utilize Soft Deletes (deleted_at, deleted_by, delete_reason) rather than hard SQL DELETE operations. This ensures complete data recovery, prevents orphaned foreign key references, and maintains historical data integrity for university compliance and auditing.

2. Enterprise File Storage Strategy (Metadata DB + Secure Filesystem):

Uploaded examination documents are NEVER stored directly in the database as BLOBs. Storing binary documents as BLOBs causes database bloat, severely degrades backup/restore performance, impairs memory caching (RAM), and prevents efficient file streaming. Instead, encrypted documents reside on a secure, restricted server filesystem while the database strictly stores file metadata (Original File Name, Stored File Name, Storage Path, File Hash SHA-256, File Size, MIME Type, Upload Timestamp, Uploaded By).

3. Decoupled Cross-Department Lecturer Assignments (lecturer_course_assignments):

Lecturer allocations are decoupled from users and courses tables into a dedicated junction table. Lecturers can teach single, multiple, or cross-departmental courses across any of FCIT's 5 departments (CSC, SWE, CYB, DAT, ICT) bound to specific academic sessions, semesters, and levels.

4. Flexible Department & Level-Based Moderation (moderator_level_assignments):

Moderators are assigned by BOTH Department, Academic Level (100L-400L), Academic Session, and Semester. A single moderator can handle multiple levels and departments, while different departments can assign distinct moderators for the same level without data redundancy.

5. Immutable Paper Versioning (paper_versions):

Examination papers are NEVER overwritten. Every revision or post-moderation update creates a completely new version entry with incremental version numbering, change summaries, and SHA-256 file checksums.

6. Dedicated Paper Status Lifecycle History (paper_status_history):

Paper state transitions (Draft -> Submitted -> Returned for Correction -> Resubmitted -> Approved -> Blind Lockdown -> Printed -> Archived) are stored in an append-only status history table recording Previous Status, New Status, Changed By, Change Timestamp, and Optional Remarks.

7. Decoupled Two-Stage Secure Printing (print_requests & print_jobs):

Separates print governance into stage 1 print_requests (Pending, Approved, Rejected) and stage 2 print_jobs (generated ONLY from approved requests).

8. Time-Enforced Examination Schedules (examination_schedules):

Maintains official timetable dates, times, and venues. Used by application business logic to strictly restrict Exam Officer printing to exactly one day (24 hours) before the scheduled examination.

9. Full Encryption Metadata Envelope (blind_lockdowns):

Encapsulates server-side AES-256 encryption metadata: Encryption Algorithm, Key Identifier, Encryption Status, Lock Timestamp, Locked By, Lock Reason, Unlock Timestamp, and Author Access Revocation flags.

10. Dynamic Watermark Audit Logs (watermark_logs):

Logs dynamic watermark generation per viewer session, user, and document for digital forensics.

11. Independent Authentication Log (login_history):

Isolates authentication logging (Successful/Failed logins, IP, Browser, Device, OS, Login/Logout Timestamps) from operational transaction audit logs to optimize security monitoring and brute-force tracking.

12. Centralized 6-Role Notification Pipeline (notifications):

A unified messaging table serving all system roles (System Administrator, HOD, Exam Officer, Moderator, Lecturer, Reserved) with full sender, recipient, and read tracking.

13. Unified Enterprise Audit Log (audit_logs):

Single append-only audit trail capturing all CRUD transactions and administrative actions across all system entities.

2. REFINED DATABASE DIRECTORY & TABLE CATALOG (37 TABLES)

Domain 1: Organization & Academic Calendar

- departments : Master list of the 5 FCIT academic departments (Supports Soft Delete).
- academic_sessions : Academic calendar years (e.g., 2025/2026).
- semesters : Term breakdown (Harmattan / Rain).

Domain 2: Identity, RBAC & Assignments

- users : Core identity repository for all system actors (Supports Soft Delete).
- roles : System roles catalog (Admin, HOD, Exam Officer, Moderator, Lecturer, Reserved).
- user_roles : Junction table mapping users to security roles.
- hod_assignments : Departmental HOD assignments (Strict 1 active HOD per department).
- exam_officer_assignments : Departmental Exam Officer assignments (Strict 1 active EO per department).
- moderator_level_assignments : Maps Moderators by Department, Academic Level, Session, and Semester.
- lecturer_course_assignments : Dedicated junction table for cross-departmental lecturer allocations.
- hod_delegations : Temporary acting HOD delegations.

Domain 3: Curriculum & Examination Timetable

- courses : Master listing of departmental courses and unit loads (Supports Soft Delete).
- examination_schedules : Official exam timetable enforcing the 24-hr print window.

Domain 4: Examination Paper Lifecycle & Review Pipeline

- examination_papers : Master entity for exam papers (Supports Soft Delete).
- paper_versions : Immutable paper versions (Filesystem metadata + version history).
- paper_status_history : Dedicated audit record of every workflow status transition.
- review_comments : Threaded peer-review comments between Moderator and Lecturer.
- approval_records : Verification certificate and SHA-256 signature generated upon approval.

Domain 5: Encryption Metadata & Blind Lockdown

- blind_lockdowns : Server-side encryption envelope, algorithm, key ID, and lock/unlock metadata.
- lockdown_history : Audit trail of lock and emergency unlock requests.
- encryption_keys : Encrypted envelope key references (Master Key wrapped).

Domain 6: Secure Viewer (DRM)

- viewer_sessions : Active in-browser DRM viewing session tracking.
- watermark_logs : Dynamic watermark rendering log (Watermark Session, User, Document, Viewer Session).
- screen_protection_logs : Captured screen protection violations (PrintScreen, Ctrl+P, hotkeys).

Domain 7: Secure Printing Pipeline

- printer_devices : Authorized physical network printer registry.
- print_requests : Stage 1: Formal print requests (Pending, Approved, Rejected).
- print_jobs : Stage 2: Generated print execution jobs derived ONLY from approved requests.
- print_logs : Stage 3: Immutable post-print execution audit log.

Domain 8: Audit & Authentication

- audit_logs : Centralized enterprise audit trail across all system entities.
- login_history : Dedicated authentication log (Success, Failed, IP, OS, Device, Timestamps).
- security_events : High-priority security exception and intrusion log.

Domain 9: Messaging & Communication

- notifications : Centralized notification store serving all 6 user roles.

Domain 10: Enterprise Archival Systems

- archived_papers : Cold storage archive for finalized examination papers.
- archived_audit_logs : Historical transaction logs moved out of hot production tables.
- archived_print_logs : Historical print operation records.
- archived_workflow_history : Historical session workflow metrics.

Domain 11: System Configuration

- system_settings : Faculty and departmental configuration parameters.

3. DETAILED REFINED TABLE SPECIFICATIONS

DOMAIN 1: ORGANIZATION & ACADEMIC CALENDAR

1. departments

- Purpose: Stores the 5 official academic departments under FCIT.

- Columns:

- id : INT, Primary Key, Auto-Increment
- code : VARCHAR(10), NOT NULL, Unique (e.g., 'CSC', 'SWE', 'CYB', 'DAT', 'ICT')

- name : VARCHAR(150), NOT NULL, Unique
- deleted_at : TIMESTAMP, NULL (Soft Delete Timestamp)
- deleted_by : INT, NULL, Foreign Key -> users(id)
- delete_reason : VARCHAR(255), NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- updated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
- Foreign Keys: deleted_by REFERENCES users(id)
- Relationships: One-to-Many with courses, users, hod_assignments, exam_officer_assignments, moderator_level_assignments, examination_schedules.
- Soft Delete Strategy: Soft deletion prevents cascading deletion of historical departmental exam records.
- Indexes: idx_dept_code (code), idx_dept_deleted (deleted_at)

2. academic_sessions

- Purpose: Defines academic calendar years (e.g., 2025/2026).

- Columns:

- id : INT, Primary Key, Auto-Increment
- session_name : VARCHAR(20), NOT NULL, Unique (e.g., '2025/2026')
- start_date : DATE, NOT NULL
- end_date : DATE, NOT NULL
- is_current : BOOLEAN, DEFAULT: FALSE
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Relationships: One-to-Many with semesters, examination_papers, examination_schedules.
- Indexes: idx_session_current (is_current)

3. semesters

- Purpose: Defines academic terms within a session.

- Columns:

- id : INT, Primary Key, Auto-Increment
- academic_session_id : INT, NOT NULL, Foreign Key -> academic_sessions(id)
- semester_name : ENUM('Harmattan', 'Rain', 'First', 'Second'), NOT NULL
- is_active : BOOLEAN, DEFAULT: FALSE
- start_date : DATE, NOT NULL
- end_date : DATE, NOT NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Foreign Keys: academic_session_id REFERENCES academic_sessions(id) ON DELETE RESTRICT
- Unique Constraints: Composite (academic_session_id, semester_name)
- Relationships: Many-to-One with academic_sessions, One-to-Many with examination_papers, examination_schedules.
- Indexes: idx_semester_active (is_active, academic_session_id)

DOMAIN 2: IDENTITY, RBAC & ASSIGNMENTS

4. users

- Purpose: Core identity repository for all system actors.

- Columns:

- id : INT, Primary Key, Auto-Increment
- staff_id : VARCHAR(50), NOT NULL, Unique
- first_name : VARCHAR(100), NOT NULL
- last_name : VARCHAR(100), NOT NULL

- email : VARCHAR(150), NOT NULL, Unique
- phone_number : VARCHAR(20), NULL
- password_hash : VARCHAR(255), NOT NULL
- primary_department_id : INT, NOT NULL, Foreign Key -> departments(id)
- status : ENUM('Active', 'Suspended', 'Deactivated'), DEFAULT: 'Active'
- force_password_change : BOOLEAN, DEFAULT: TRUE
- last_login_at : TIMESTAMP, NULL
- deleted_at : TIMESTAMP, NULL (Soft Delete Timestamp)
- deleted_by : INT, NULL, Foreign Key -> users(id)
- delete_reason : VARCHAR(255), NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- updated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
- Foreign Keys: primary_department_id REFERENCES departments(id), deleted_by REFERENCES users(id)
- Soft Delete Strategy: When a staff member leaves, soft deletion retains their user record so historical paper uploads, approvals, and audit trails remain fully intact and attributable.
- Indexes: idx_user_email (email), idx_user_staff (staff_id), idx_user_deleted (deleted_at)

5. roles

- Purpose: System roles catalog.
- Columns: id, role_code (Unique), description.

6. user_roles

- Purpose: Junction table mapping users to security roles.
- Columns: id, user_id, role_id, assigned_by, assigned_at.

7. hod_assignments

- Purpose: Manages HOD leadership assignments per department.
- Columns: id, user_id, department_id, academic_session_id, start_date, end_date, is_active, assigned_by, created_at.

8. exam_officer_assignments

- Purpose: Manages Exam Officer assignments per department.
- Columns: id, user_id, department_id, academic_session_id, start_date, end_date, is_active, assigned_by, created_at.

9. moderator_level_assignments [REFINED REQUIREMENT #4]

- Purpose: Maps Moderators by Department, Academic Level (100L-400L), Academic Session, and Semester.

- Columns:

- id : INT, Primary Key, Auto-Increment
- moderator_id : INT, NOT NULL, Foreign Key -> users(id)
- department_id : INT, NOT NULL, Foreign Key -> departments(id)
- academic_session_id : INT, NOT NULL, Foreign Key -> academic_sessions(id)
- semester_id : INT, NOT NULL, Foreign Key -> semesters(id)
- level : ENUM('100', '200', '300', '400'), NOT NULL
- status : ENUM('Active', 'Inactive'), DEFAULT: 'Active'
- assigned_by : INT, NOT NULL, Foreign Key -> users(id)
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Foreign Keys: moderator_id REFERENCES users(id), department_id REFERENCES departments(id), academic_session_id REFERENCES academic_sessions(id), semester_id REFERENCES semesters(id), assigned_by REFERENCES users(id)
- Unique Constraints: Composite (department_id, academic_session_id, semester_id, level, status)

- Business Rules & Flexibility:

1. One moderator can handle multiple levels.

2. One moderator can handle multiple departments.

3. Different departments can assign distinct moderators for the same level.

- Indexes: idx_mod_assign (department_id, academic_session_id, semester_id, level)

10. lecturer_course_assignments [REFINED REQUIREMENT #3]

- Purpose: Dedicated junction table managing Lecturer course allocations, explicitly supporting cross-departmental teaching.

- Columns:

- id : INT, Primary Key, Auto-Increment
- lecturer_id : INT, NOT NULL, Foreign Key -> users(id)
- course_id : INT, NOT NULL, Foreign Key -> courses(id)
- department_id : INT, NOT NULL, Foreign Key -> departments(id)
- academic_session_id : INT, NOT NULL, Foreign Key -> academic_sessions(id)
- semester_id : INT, NOT NULL, Foreign Key -> semesters(id)
- level : ENUM('100', '200', '300', '400'), NOT NULL
- assignment_status : ENUM('Active', 'Transferred', 'Revoked'), DEFAULT: 'Active'
- assigned_by : INT, NOT NULL, Foreign Key -> users(id)
- assigned_date : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Foreign Keys: lecturer_id REFERENCES users(id), course_id REFERENCES courses(id), department_id REFERENCES departments(id), academic_session_id REFERENCES academic_sessions(id), semester_id REFERENCES semesters(id), assigned_by REFERENCES users(id)
- Unique Constraints: Composite (lecturer_id, course_id, academic_session_id, semester_id)
- Indexes: idx_lect_assign (lecturer_id, course_id, academic_session_id)

11. hod_delegations

- Purpose: Temporary delegation of HOD responsibilities to an acting officer.
- Columns: id, delegated_by, acting_officer_id, department_id, start_date, end_date, reason, status, created_at.

DOMAIN 3: CURRICULUM & EXAMINATION TIMETABLE

12. courses

- Purpose: Repository of all FCIT departmental courses.

- Columns:

- id : INT, Primary Key, Auto-Increment
- department_id : INT, NOT NULL, Foreign Key -> departments(id)
- course_code : VARCHAR(15), NOT NULL, Unique
- course_title : VARCHAR(200), NOT NULL
- unit_load : TINYINT, NOT NULL
- level : ENUM('100', '200', '300', '400'), NOT NULL
- semester_type : ENUM('Harmattan', 'Rain', 'Both'), NOT NULL
- deleted_at : TIMESTAMP, NULL (Soft Delete Timestamp)
- deleted_by : INT, NULL, Foreign Key -> users(id)
- delete_reason : VARCHAR(255), NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- updated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

- Soft Delete Strategy: Soft deleting a retired course preserves historical exam papers.
- Indexes: idx_course_code (course_code), idx_course_deleted (deleted_at)

13. examination_schedules [REFINED REQUIREMENT #9]

- Purpose: Official examination timetable entity storing exam dates, times, and venues. Crucial for driving Blind Lockdown and Secure Printing rules.

- Columns:

- id : INT, Primary Key, Auto-Increment
- academic_session_id : INT, NOT NULL, Foreign Key -> academic_sessions(id)
- semester_id : INT, NOT NULL, Foreign Key -> semesters(id)
- department_id : INT, NOT NULL, Foreign Key -> departments(id)
- course_id : INT, NOT NULL, Foreign Key -> courses(id)
- level : ENUM('100', '200', '300', '400'), NOT NULL
- examination_date : DATE, NOT NULL
- start_time : TIME, NOT NULL
- end_time : TIME, NOT NULL
- venue : VARCHAR(150), NOT NULL
- created_by : INT, NOT NULL, Foreign Key -> users(id)
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- updated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
- Business Rules: Used by application logic to enforce the 24-hr secure print window.
- Indexes: idx_exam_sched_lookup (course_id, examination_date)

DOMAIN 4: EXAMINATION PAPER LIFECYCLE & REVIEW PIPELINE

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14. examination_papers [REFINED REQUIREMENT #10 Soft Delete]

- Purpose: Master entity for examination paper records.

- Columns:

- id : INT, Primary Key, Auto-Increment
- course_id : INT, NOT NULL, Foreign Key -> courses(id)
- department_id : INT, NOT NULL, Foreign Key -> departments(id)
- academic_session_id : INT, NOT NULL, Foreign Key -> academic_sessions(id)
- semester_id : INT, NOT NULL, Foreign Key -> semesters(id)
- level : ENUM('100', '200', '300', '400'), NOT NULL
- created_by : INT, NOT NULL, Foreign Key -> users(id)
- assigned_moderator_id : INT, NULL, Foreign Key -> users(id)
- examination_schedule_id: INT, NULL, Foreign Key -> examination_schedules(id)
- current_status : ENUM('Draft', 'Submitted', 'Returned for Correction', 'Resubmitted', 'Approved', 'Blind Lockdown', 'Printed', 'Archived'), DEFAULT: 'Draft'
- current_version_number: INT, DEFAULT: 1
- deleted_at : TIMESTAMP, NULL (Soft Delete Timestamp)
- deleted_by : INT, NULL, Foreign Key -> users(id)
- delete_reason : VARCHAR(255), NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- updated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
- Indexes: idx_paper_status (department_id, current_status), idx_paper_deleted (deleted_at)

15. paper_versions [REFINED REQUIREMENT #2 & #11 File Storage Strategy]

- Purpose: Dedicated entity ensuring complete version history of examination papers. Files are NEVER overwritten.

- Columns:

- id : INT, Primary Key, Auto-Increment
- paper_id : INT, NOT NULL, Foreign Key -> examination_papers(id) ON DELETE CASCADE
- version_number : INT, NOT NULL
- original_file_name : VARCHAR(255), NOT NULL
- stored_file_name : VARCHAR(255), NOT NULL
- storage_path : VARCHAR(255), NOT NULL
- file_hash : VARCHAR(64), NOT NULL (SHA-256)
- file_size_bytes : INT, NOT NULL
- mime_type : VARCHAR(100), NOT NULL
- marking_guide_path : VARCHAR(255), NULL
- marking_guide_hash : VARCHAR(64), NULL
- change_summary : TEXT, NULL
- uploaded_by : INT, NOT NULL, Foreign Key -> users(id)
- upload_timestamp : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Indexes: idx_version_lookup (paper_id, version_number)

16. paper_status_history [REFINED REQUIREMENT #1]

- Purpose: Dedicated Paper Status History entity recording every workflow transition.

- Columns:

- id : INT, Primary Key, Auto-Increment
- paper_id : INT, NOT NULL, Foreign Key -> examination_papers(id) ON DELETE CASCADE
- paper_version_id : INT, NULL, Foreign Key -> paper_versions(id)
- previous_status : VARCHAR(50), NULL
- new_status : VARCHAR(50), NOT NULL
- changed_by : INT, NOT NULL, Foreign Key -> users(id)
- change_timestamp : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- remarks : TEXT, NULL
- Indexes: idx_status_hist_paper (paper_id, change_timestamp)

17. review_comments

- Purpose: Peer-review comment threads between Moderator and Lecturer.
- Columns: id, paper_id, paper_version_id, author_id, comment_type, section_reference, comment_text, parent_id, is_resolved, created_at.

18. approval_records

- Purpose: Digital approval certificate generated upon moderator approval.
- Columns: id, paper_id (Unique), paper_version_id, moderator_id, department_id, approval_certificate_id (Unique), approval_hash, approval_date, remarks.

DOMAIN 5: ENCRYPTION METADATA & BLIND LOCKDOWN PROTOCOL

19. blind_lockdowns [REFINED REQUIREMENT #6 Encryption Metadata]

- Purpose: Encryption Metadata entity encapsulating server-side AES-256 encryption metadata and Blind Lockdown state.

- Columns:

- id : INT, Primary Key, Auto-Increment
- paper_id : INT, NOT NULL, Unique, Foreign Key -> examination_papers(id) ON DELETE CASCADE
- paper_version_id : INT, NOT NULL, Foreign Key -> paper_versions(id)
- encryption_algorithm : VARCHAR(50), DEFAULT: 'AES-256-CBC'

- key_identifier : VARCHAR(100), NOT NULL, Unique
- encryption_status : ENUM('Pending', 'Encrypted', 'Emergency Unlocked', 'Archived'), DEFAULT: 'Pending'
- lock_timestamp : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- locked_by : INT, NULL, Foreign Key -> users(id)
- lock_reason : VARCHAR(255), DEFAULT: 'AUTOMATIC_MODERATION_APPROVAL_LOCKDOWN'
- unlock_timestamp : TIMESTAMP, NULL
- unlocked_by : INT, NULL, Foreign Key -> users(id)
- unlock_reason : TEXT, NULL
- author_access_revoked : BOOLEAN, DEFAULT: TRUE
- Indexes: idx_lockdown_paper (paper_id), idx_lockdown_status (encryption_status)

20. lockdown_history

- Purpose: Audit trail of lock and emergency unlock requests.
- Columns: id, blind_lockdown_id, paper_id, action_type, performed_by, reason, ip_address, action_timestamp.

21. encryption_keys

- Purpose: Wrapped key references for AES-256 decryption.
- Columns: id, paper_id (Unique), wrapped_key_reference, key_fingerprint, created_at.

DOMAIN 6: SECURE VIEWER & DRM SESSION TRACKING

22. viewer_sessions

- Purpose: Active in-browser DRM viewing sessions.
- Columns: id, paper_id, user_id, session_token (Unique), ip_address, user_agent, session_start, session_end, is_active.

23. watermark_logs

- Purpose: Audit record of dynamic watermarks rendered during viewing sessions.
- Columns: id, watermark_session_id, viewer_session_id, user_id, paper_id, watermark_text, rendered_timestamp.

24. screen_protection_logs

- Purpose: Logs screen capture / DRM key violations during viewing sessions.
- Columns: id, viewer_session_id, user_id, violation_type, captured_at.

DOMAIN 7: SECURE PRINTING PIPELINE (DECOUPLED REQUEST & JOB MODEL)

25. printer_devices

- Purpose: Registry of authorized network printers for paper output.
- Columns: id, department_id, device_name, ip_address (Unique), mac_address (Unique), location_description, is_active, created_at.

26. print_requests [REFINED REQUIREMENT #5 Decoupled Secure Printing]

- Purpose: Stage 1 Print Request entity created by Exam Officers. Must be explicitly approved or rejected.

- Columns:

- id : INT, Primary Key, Auto-Increment
- paper_id : INT, NOT NULL, Foreign Key -> examination_papers(id)
- examination_schedule_id: INT, NOT NULL, Foreign Key -> examination_schedules(id)
- requested_by : INT, NOT NULL, Foreign Key -> users(id)

- copies_requested : SMALLINT, NOT NULL
- justification : TEXT, NULL
- status : ENUM('Pending', 'Approved', 'Rejected'), DEFAULT: 'Pending'
- reviewed_by : INT, NULL, Foreign Key -> users(id)
- review_remarks : TEXT, NULL
- reviewed_at : TIMESTAMP, NULL
- created_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- Indexes: idx_print_req_status (status), idx_print_req_paper (paper_id)

27. print_jobs [REFINED REQUIREMENT #5 Decoupled Secure Printing]

- Purpose: Stage 2 Print Job entity generated ONLY from approved print requests.

- Columns:

- id : INT, Primary Key, Auto-Increment
- print_request_id : INT, NOT NULL, Unique, Foreign Key -> print_requests(id)
- paper_id : INT, NOT NULL, Foreign Key -> examination_papers(id)
- printer_device_id : INT, NOT NULL, Foreign Key -> printer_devices(id)
- assigned_exam_officer_id: INT, NOT NULL, Foreign Key -> users(id)
- copies_to_print : SMALLINT, NOT NULL
- job_status : ENUM('Ready', 'Printing', 'Completed', 'Aborted'), DEFAULT: 'Ready'
- generated_at : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- completed_at : TIMESTAMP, NULL
- Indexes: idx_pjob_status (job_status)

28. print_logs

- Purpose: Stage 3: Immutable execution log of completed print operations.
- Columns: id, print_job_id, paper_id, user_id, printer_device_id, department_id, copies_printed, ip_address, user_agent, status, error_message, created_at.

DOMAIN 8: AUDIT & AUTHENTICATION

29. audit_logs

- Purpose: Centralized enterprise audit trail.
- Columns: id, user_id, staff_id, role_code, department_code, action, entity_name, entity_id, old_values, new_values, description, ip_address, user_agent, created_at.

30. login_history [REFINED REQUIREMENT #7 Independent Login History]

- Purpose: Dedicated Login History entity strictly isolated from the general audit log.

- Columns:

- id : BIGINT, Primary Key, Auto-Increment
- user_id : INT, NULL, Foreign Key -> users(id) ON DELETE SET NULL
- attempted_identifier : VARCHAR(150), NOT NULL
- login_status : ENUM('Successful_Login', 'Failed_Password', 'User_NotFound', 'Account_Locked'), NOT NULL
- ip_address : VARCHAR(45), NOT NULL
- browser : VARCHAR(100), NULL
- device : VARCHAR(50), NULL
- operating_system : VARCHAR(100), NULL
- login_timestamp : TIMESTAMP, DEFAULT: CURRENT_TIMESTAMP
- logout_timestamp : TIMESTAMP, NULL

• Indexes: idx_login_user_time (user_id, login_timestamp), idx_login_ip_status (ip_address, login_status)

31. security_events

• Purpose: Intrusion alerts and high-severity security exception logs.
• Columns: id, event_type, severity, description, user_id, ip_address, browser_fingerprint, created_at.

DOMAIN 9: MESSAGING & COMMUNICATION

32. notifications [REFINED REQUIREMENT #8 Centralized Notification System]

• Purpose: Single centralized Notifications entity supporting all 6 user roles.
• Columns: id, sender_id, recipient_id, notification_type, title, message, link_url, read_status, timestamp.
• Indexes: idx_notif_recipient (recipient_id, read_status)

DOMAIN 10: ENTERPRISE ARCHIVAL SYSTEMS

33. archived_papers

34. archived_audit_logs

35. archived_print_logs

36. archived_workflow_history

DOMAIN 11: SYSTEM CONFIGURATION

37. system_settings

• Purpose: Global and departmental configuration parameters.
• Columns: id, department_id, setting_key, setting_value, description, updated_by, updated_at.
• Indexes: idx_settings_key (department_id, setting_key)

4. FINAL RELATIONSHIP VALIDATION SUMMARY

A. One-to-One (1:1) Relationships:

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1. examination_papers <-> approval_records

2. examination_papers <-> blind_lockdowns

3. examination_papers <-> encryption_keys

4. print_requests <-> print_jobs

B. One-to-Many (1:N) Relationships:

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1. departments -> courses

2. academic_sessions -> semesters

3. examination_papers -> paper_versions

4. examination_papers -> paper_status_history

5. paper_versions -> review_comments

6. viewer_sessions -> watermark_logs

7. viewer_sessions -> screen_protection_logs

8. print_requests -> print_jobs

9. print_jobs -> print_logs

10. users -> audit_logs

11. users -> login_history

12. users -> notifications

13. examination_schedules -> examination_papers

C. Many-to-Many (N:M) Relationships:

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1. users <-> roles (via user_roles)

2. users (Lecturers) <-> courses (via lecturer_course_assignments)

3. users (Moderators) <-> departments / levels (via moderator_level_assignments)

5. SUMMARY OF REFINED ENTERPRISE COMPLIANCE (ALL 12 POINTS)

Point #1 : Paper Status History -> paper_status_history entity tracking status transitions.

Point #2 : Paper Versioning -> paper_versions entity preserving all drafts/revisions.

Point #3 : Lecturer Course Assignments -> lecturer_course_assignments dedicated junction table.

Point #4 : Moderator Assignments -> moderator_level_assignments mapped by Department, Level, Session, Semester.

Point #5 : Secure Printing -> Decoupled print_requests and print_jobs.

Point #6 : Encryption Metadata -> blind_lockdowns table with full metadata.

Point #7 : Login History -> login_history entity isolated from general audit log.

Point #8 : Notification System -> Centralized notifications entity serving all 6 roles.

Point #9 : Examination Schedule -> examination_schedules entity enforcing 24-hr print rule.

Point #10 : Soft Delete Strategy -> deleted_at, deleted_by, delete_reason fields included.

Point #11 : File Storage Strategy -> Metadata DB + Secure Filesystem strategy mandated.

Point #12 : Final Relationship Validation-> Comprehensive summary of 1:1, 1:N, N:M relationships provided.

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6. DATABASE DESIGN STANDARDS AND CONVENTIONS

This section defines the implementation standards and conventions that MUST be strictly adhered to during SQL DDL script generation, migration development, and application persistence operations.

1. Naming Conventions:

- Table Names: Lowercase, snake_case, plural nouns (e.g., `users`, `departments`, `examination_papers`, `paper_versions`, `audit_logs`).
- Column Names: Lowercase, snake_case, singular descriptive names (e.g., `staff_id`, `first_name`, `email`, `file_hash`).
- Primary Keys: Standardized to `id` across all tables.
- Foreign Keys: Singular parent table name + `_id` suffix (e.g., `user_id`, `department_id`, `course_id`, `paper_id`).
- Index Names: Prefix `idx_{table_short}_{column_name}` for standard indexes (e.g., `idx_user_email`, `idx_paper_status`).
- Unique Constraint Names: Prefix `uk_{table_short}_{column_name}` for unique constraints (e.g., `uk_user_staff_id`, `uk_dept_code`).
- Foreign Key Constraint Names: Prefix `fk_{source_table}_{target_table}_{column}` (e.g., `fk_papers_courses_course_id`).

2. Standard Timestamp Fields:

- `created_at`: TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP. Records exact entity creation time.
- `updated_at`: TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP. Automatically updates on row modification.
- `deleted_at`: TIMESTAMP NULL. Stores soft-delete timestamp (NULL indicates active, non-deleted row).

3. Primary Key Strategy:

- Transactional & Domain Master Entities: `INT UNSIGNED AUTO_INCREMENT PRIMARY KEY` (supports up to 4.29 billion records per entity).
- High-Volume Audit & Security Logs (`audit_logs`, `login_history`, `security_events`): `BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY` (supports up to 18.4 quintillion records).

4. Foreign Key Naming Conventions:

- Foreign key constraints must explicitly specify constraint names following the pattern `fk_{child_table}_{parent_table}_{key}`.
- Every foreign key column must match the data type and signedness of its referenced primary key (e.g., `INT UNSIGNED` to `INT UNSIGNED`).

5. Indexing Strategy:

- Primary Keys: Automatically indexed by InnoDB cluster key.
- Foreign Keys: ALL foreign key columns must have an explicit B-Tree index to optimize JOIN performance and prevent full table scans.
- Status & Workflow Flags: Index status columns frequently filtered in dashboard queries (e.g., `status`, `is_active`, `is_current`).
- Unique Natural Keys: Unique indexes created for natural keys (`staff_id`, `email`, `course_code`, `approval_certificate_id`).
- Soft Delete Flags: Compound index on `(deleted_at)` or `(id, deleted_at)` to accelerate active record filtering.

6. Character Set and Collation:

- Character Set: `utf8mb4` (Full 4-byte UTF-8 support including special characters and symbols).
- Collation: `utf8mb4_unicode_ci` (Unicode-compliant, case-insensitive, accurate accent sorting).

7. Storage Engine:

- Engine: `InnoDB` across ALL tables.
- Justification: Provides full ACID compliance, row-level locking, crash recovery, foreign key constraint enforcement, and high concurrency performance under enterprise workloads.

8. Referential Integrity Rules:

- Parent Delete Restrictions (`ON DELETE RESTRICT`): Applied to core reference tables (e.g., deleting a department or course is restricted if active papers or users reference it).
- Cascading Deletes (`ON DELETE CASCADE`): Applied ONLY to dependent lifecycle child tables (e.g., deleting an exam paper cascades to delete its `paper_versions`, `paper_status_history`, and `review_comments`).
- Nullification (`ON DELETE SET NULL`): Applied to optional audit/user references (e.g., `user_id` in audit logs becomes NULL if a user is purged, preserving audit integrity).
- Update Rules (`ON UPDATE CASCADE`): Ensures foreign key references automatically update if a parent primary key changes.

9. File Storage Conventions:

- Filesystem Storage: Uploaded examination PDFs/DOCXs are stored in a secure, non-web-accessible server storage directory (`/storage/secure_papers/`).
- Database Storage: The database stores ONLY metadata (`original_file_name`, `stored_file_name`, `storage_path`, `file_hash`, `file_size_bytes`, `mime_type`).
- Document BLOBs are strictly forbidden in the database schema.

10. Security & Confidentiality Conventions:

- Password Hashing: Hashes must be generated via Argon2id or bcrypt (stored in `password_hash` VARCHAR(255)).
- Audit Immutability: `audit_logs` and `login_history` tables are append-only. Application DB user accounts lack UPDATE and DELETE privileges on audit tables.
- Soft Delete Enforcement: Soft deletion (`deleted_at`, `deleted_by`) preserves record history for forensic and administrative review.

- Encryption Envelope: AES-256 encryption metadata (`encryption_algorithm`, `key_identifier`, `encryption_status`, `author_access_revoked`) is stored alongside paper records.

11. Data Validation & Constraint Rules:

- Required Fields: Critical columns are declared `NOT NULL`.
- Unique Constraints: Enforced at database layer for staff IDs, emails, course codes, session names, printer IPs, and certificates.
- Sensible Defaults: Status defaults set explicitly (`status = 'Active'`, `is_read = FALSE`, `author_access_revoked = TRUE`).

END OF ENTERPRISE DATABASE BLUEPRINT (REFINED VERSION 2.0)